www.tntextbooks.in

To create a digital signature, signing software (email) creates a one-way hash of the electronic data to be signed. The user's private key to encrypt the hash, returning a value that is unique to the hashed data. The encrypted hash, along with other information such as the hashing algorithm, forms the digital signature. Any change in the data, even to a single bit, results in a different hash value. This attribute enables others to validate the integrity of the data by using the signer's public key to decrypt the hash. If the decrypted hash matches a second computed hash of the same data, it proves that the data hasn't changed since it was signed. If the two hashes don't match, the data has either been tampered with in some way (indicating a failure of integrity) or the signature was created with a private key that doesn't correspond to the public key presented by the signer (indicating a failure of authentication). **Figure 17.12** shows the function of a digital signature.

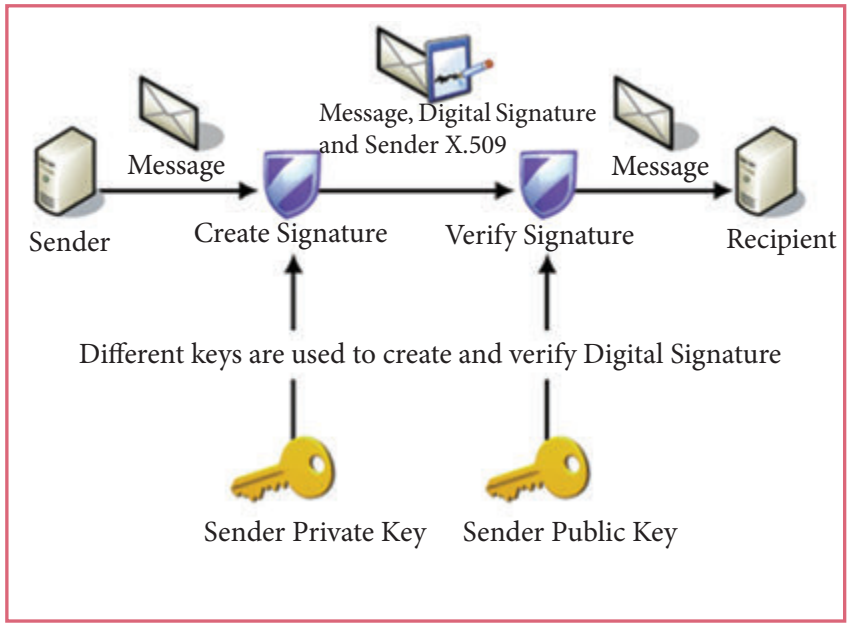

Figure 17.12 –Function of Digital Signature

17.4 INTRODUCTION TO INFORMATION TECHNOLOGY ACT

In the 21stcentury, Computer, Internet and ICT or e-revolution has changed the life style of the people. Today paper based communication has been substituted by e-communication. Accordingly we have new terminologies like cyber world, e-transaction, e-banking, e-return and e-contracts. Apart from positive side of e-revolution there is also negative side of computer, that is, the internet and ICT in the hands of criminals which has become a weapon of offence. Accordingly a new panel of members emerged to tackle the problems of cyber crimes in cyber space i.e. Cyber Law or Cyber Space Law or Information Technology Law or Internet Law.

In India Cyber law and IT Act 2000 , modified in 2008 are being articulated to prevent computer crimes. IT Act 2000 is an act to provide legal recognition for transactions carried out by means of **ElectronicData Interchange(EDI)** and other means of electronic communication. It is the primary law in India dealing with cybercrime and electronic commerce(e-Commerce). e-Commerce is electronic data exchange or electronic filing of information.